

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	The velocity of the ball has reduced so KE is decreased as well as potential energy (1) By conservation of energy – energy lost against air resistance (1) Alternative: Both horizontal components of velocity calculated i.e. $36 \cos 3^\circ$ and $33 \cos 7^\circ$ ($= 35.95 \text{ m s}^{-1}$ and 32.75 m s^{-1} respectively) (1) Horizontal component of velocity has decreased so air resistance is present (1)		2		2		
		(ii)	Both vertical components calculated correctly i.e. $36 \sin 3^\circ = 1.89 \text{ [m s}^{-1}\text{]}$ and $33 \sin 7^\circ = 4.02 \text{ [m s}^{-1}\text{]}$ (1) Substitute values into a valid equation of motion e.g. $\frac{v^2 - u^2}{2ax}$ i.e. $\frac{4.02^2 - 1.89^2}{2 \times 0.75}$ (1) Valid conclusion (1) e.g. acceleration $= 8.39 \text{ [m s}^{-2}\text{]}$ so not equal to g or Height $= 0.64 \text{ [m]}$ (with $g = 9.81 \text{ m s}^{-2}$) so not equal to 0.75 [m] Or vertical velocity $= 4.22 \text{ [m s}^{-1}\text{]}$ (with $g = 9.81 \text{ m s}^{-2}$) so not equal to $4.02 \text{ [m s}^{-1}\text{]}$			3	3	3	
		(iii)	A lift force [is created by a spinning ball] (1) can be implied Due to pressure or velocity <u>difference</u> caused by the spinning or equivalent explanation (1)	2			2		
			Question 9 total	6	9	5	20	10	0